



1. Description

1.1. Project

Project Name	STM32h753bit6
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	04/29/2026

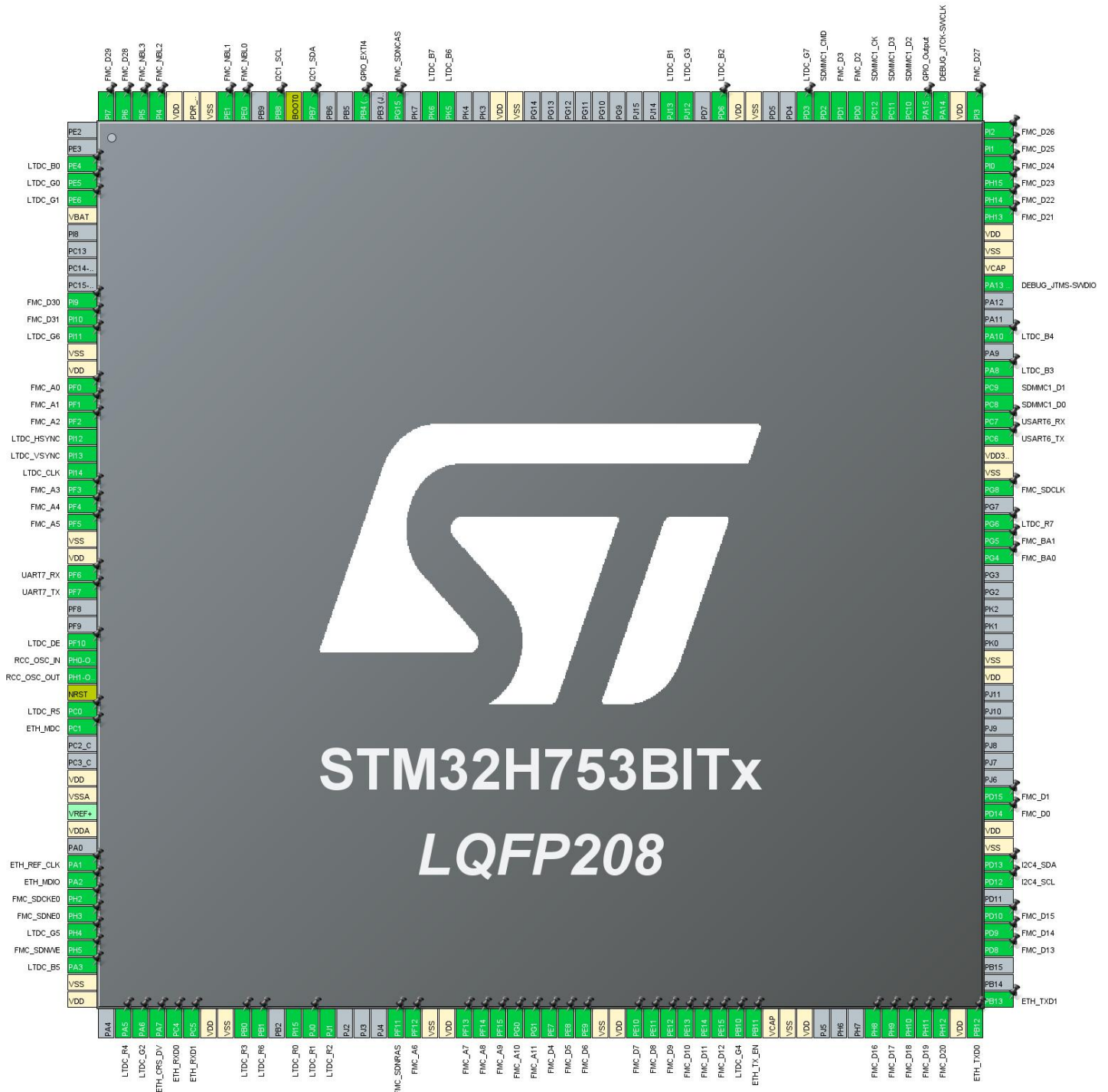
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H743/753
MCU name	STM32H753BITx
MCU Package	LQFP208
MCU Pin number	208

1.3. Core(s) information

Core(s)	ARM Cortex-M7
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
3	PE4	I/O	LTDC_B0	
4	PE5	I/O	LTDC_G0	
5	PE6	I/O	LTDC_G1	
6	VBAT	Power		
11	PI9	I/O	FMC_D30	
12	PI10	I/O	FMC_D31	
13	PI11	I/O	LTDC_G6	
14	VSS	Power		
15	VDD	Power		
16	PF0	I/O	FMC_A0	
17	PF1	I/O	FMC_A1	
18	PF2	I/O	FMC_A2	
19	PI12	I/O	LTDC_HSYNC	
20	PI13	I/O	LTDC_VSYNC	
21	PI14	I/O	LTDC_CLK	
22	PF3	I/O	FMC_A3	
23	PF4	I/O	FMC_A4	
24	PF5	I/O	FMC_A5	
25	VSS	Power		
26	VDD	Power		
27	PF6	I/O	UART7_RX	
28	PF7	I/O	UART7_TX	
31	PF10	I/O	LTDC_DE	
32	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
33	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
34	NRST	Reset		
35	PC0	I/O	LTDC_R5	
36	PC1	I/O	ETH_MDC	
39	VDD	Power		
40	VSSA	Power		
42	VDDA	Power		
44	PA1	I/O	ETH_REF_CLK	
45	PA2	I/O	ETH_MDIO	
46	PH2	I/O	FMC_SDCKE0	
47	PH3	I/O	FMC_SDNE0	
48	PH4	I/O	LTDC_G5	

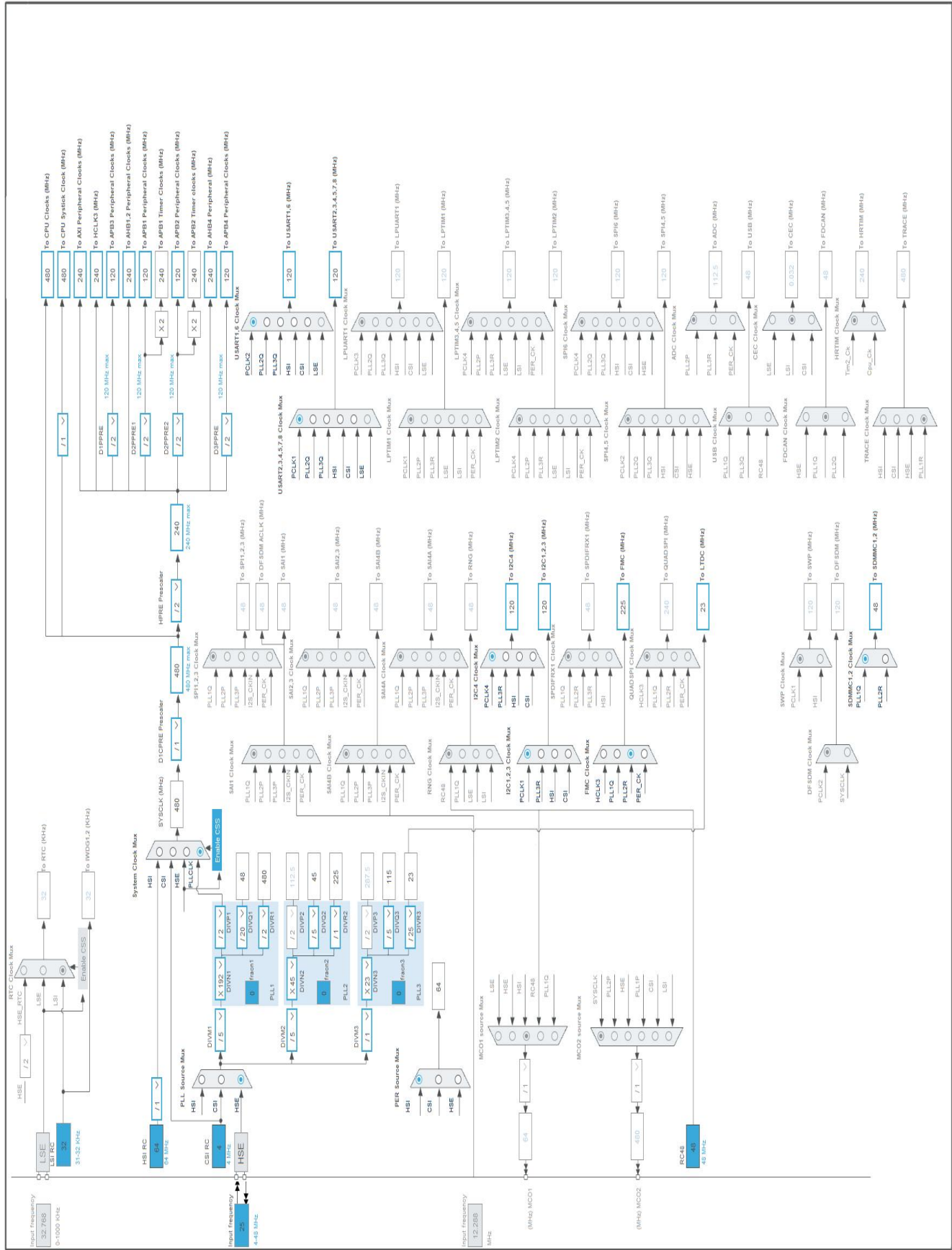
Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
49	PH5	I/O	FMC_SDNWE	
50	PA3	I/O	LTDC_B5	
51	VSS	Power		
52	VDD	Power		
54	PA5	I/O	LTDC_R4	
55	PA6	I/O	LTDC_G2	
56	PA7	I/O	ETH_CRS_DV	
57	PC4	I/O	ETH_RXD0	
58	PC5	I/O	ETH_RXD1	
59	VDD	Power		
60	VSS	Power		
61	PB0	I/O	LTDC_R3	
62	PB1	I/O	LTDC_R6	
64	PI15	I/O	LTDC_R0	
65	PJ0	I/O	LTDC_R1	
66	PJ1	I/O	LTDC_R2	
70	PF11	I/O	FMC_SDNRAS	
71	PF12	I/O	FMC_A6	
72	VSS	Power		
73	VDD	Power		
74	PF13	I/O	FMC_A7	
75	PF14	I/O	FMC_A8	
76	PF15	I/O	FMC_A9	
77	PG0	I/O	FMC_A10	
78	PG1	I/O	FMC_A11	
79	PE7	I/O	FMC_D4	
80	PE8	I/O	FMC_D5	
81	PE9	I/O	FMC_D6	
82	VSS	Power		
83	VDD	Power		
84	PE10	I/O	FMC_D7	
85	PE11	I/O	FMC_D8	
86	PE12	I/O	FMC_D9	
87	PE13	I/O	FMC_D10	
88	PE14	I/O	FMC_D11	
89	PE15	I/O	FMC_D12	
90	PB10	I/O	LTDC_G4	
91	PB11	I/O	ETH_TX_EN	
92	VCAP	Power		

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
93	VSS	Power		
94	VDD	Power		
98	PH8	I/O	FMC_D16	
99	PH9	I/O	FMC_D17	
100	PH10	I/O	FMC_D18	
101	PH11	I/O	FMC_D19	
102	PH12	I/O	FMC_D20	
103	VDD	Power		
104	PB12	I/O	ETH_TXD0	
105	PB13	I/O	ETH_TXD1	
108	PD8	I/O	FMC_D13	
109	PD9	I/O	FMC_D14	
110	PD10	I/O	FMC_D15	
112	PD12	I/O	I2C4_SCL	
113	PD13	I/O	I2C4_SDA	
114	VSS	Power		
115	VDD	Power		
116	PD14	I/O	FMC_D0	
117	PD15	I/O	FMC_D1	
124	VDD	Power		
125	VSS	Power		
131	PG4	I/O	FMC_BA0	
132	PG5	I/O	FMC_BA1	
133	PG6	I/O	LTDC_R7	
135	PG8	I/O	FMC_SDCLK	
136	VSS	Power		
137	VDD33_USB	Power		
138	PC6	I/O	USART6_TX	
139	PC7	I/O	USART6_RX	
140	PC8	I/O	SDMMC1_D0	
141	PC9	I/O	SDMMC1_D1	
142	PA8	I/O	LTDC_B3	
144	PA10	I/O	LTDC_B4	
147	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
148	VCAP	Power		
149	VSS	Power		
150	VDD	Power		
151	PH13	I/O	FMC_D21	
152	PH14	I/O	FMC_D22	

Pin Number LQFP208	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
153	PH15	I/O	FMC_D23	
154	PI0	I/O	FMC_D24	
155	PI1	I/O	FMC_D25	
156	PI2	I/O	FMC_D26	
157	PI3	I/O	FMC_D27	
158	VDD	Power		
159	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
160	PA15 (JTDI) *	I/O	GPIO_Output	
161	PC10	I/O	SDMMC1_D2	
162	PC11	I/O	SDMMC1_D3	
163	PC12	I/O	SDMMC1_CK	
164	PD0	I/O	FMC_D2	
165	PD1	I/O	FMC_D3	
166	PD2	I/O	SDMMC1_CMD	
167	PD3	I/O	LTDC_G7	
170	VSS	Power		
171	VDD	Power		
172	PD6	I/O	LTDC_B2	
174	PJ12	I/O	LTDC_G3	
175	PJ13	I/O	LTDC_B1	
184	VSS	Power		
185	VDD	Power		
188	PK5	I/O	LTDC_B6	
189	PK6	I/O	LTDC_B7	
191	PG15	I/O	FMC_SDNCAS	
193	PB4 (NJTRST)	I/O	GPIO_EXTI4	
196	PB7	I/O	I2C1_SDA	
197	BOOT0	Boot		
198	PB8	I/O	I2C1_SCL	
200	PE0	I/O	FMC_NBL0	
201	PE1	I/O	FMC_NBL1	
202	VSS	Power		
203	PDR_ON	Power		
204	VDD	Power		
205	PI4	I/O	FMC_NBL2	
206	PI5	I/O	FMC_NBL3	
207	PI6	I/O	FMC_D28	
208	PI7	I/O	FMC_D29	

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H743/753
MCU	STM32H753BITx
Datasheet	DS12117 Rev7

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

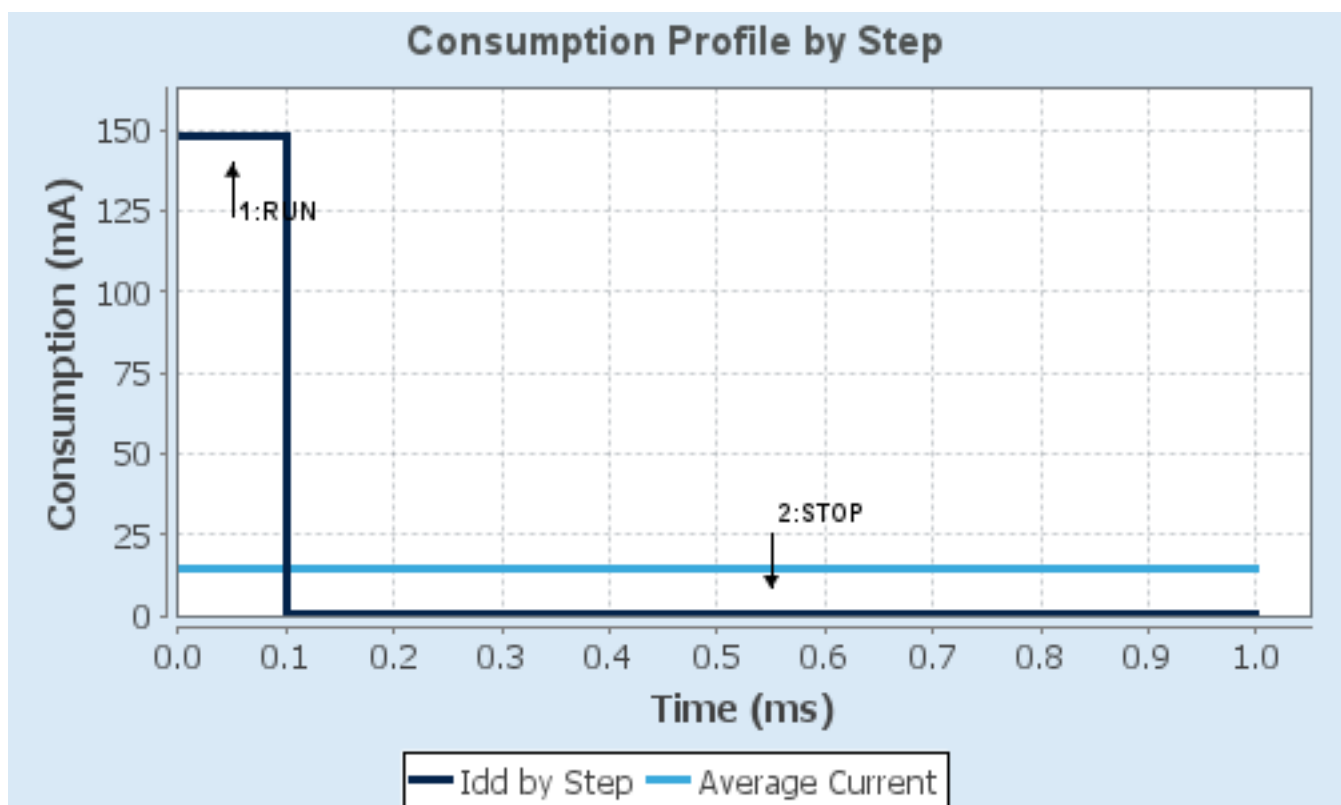
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	105.91	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	STM32h753bit6
Project Folder	C:\STM32H7\STM32h753bit6
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_H7 V1.12.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x4000
Minimum Stack Size	0x4000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	MX_GPIO_Init	GPIO
2	MX_MDMA_Init	MDMA
3	SystemClock_Config	RCC
4	MX_DMA_Init	DMA
5	MX_SDMMC1_SD_Init	SDMMC1
6	MX_FMC_Init	FMC
7	MX_I2C1_Init	I2C1
8	MX_I2C4_Init	I2C4
9	MX_UART7_Init	UART7
10	MX_USART6_UART_Init	USART6
11	MX_ETH_Init	ETH

Rank	Function Name	Peripheral Instance Name
12	MX_FATFS_Init	FATFS
13	MX_LTDC_Init	LTDC
14	MX_TIM5_Init	TIM5

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7

3.1.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode **Enabled ***

Cortex Interface Settings:

CPU ICache **Enabled ***

CPU DCache **Enabled ***

Cortex Memory Protection Unit Control Settings:

MPU Control Mode Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region Enabled
MPU Region Base Address **0x0 ***
MPU Region Size 4GB
MPU SubRegion Disable **0x87 ***
MPU TEX field level level 0
MPU Access Permission ALL ACCESS NOT PERMITTED
MPU Instruction Access DISABLE
MPU Shareability Permission ENABLE
MPU Cacheable Permission DISABLE
MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region **Enabled ***
MPU Region Base Address **0x24000000 ***
MPU Region Size **512KB ***
MPU SubRegion Disable **0x0 ***
MPU TEX field level **level 1 ***
MPU Access Permission **ALL ACCESS PERMITTED ***
MPU Instruction Access **DISABLE ***
MPU Shareability Permission DISABLE
MPU Cacheable Permission **ENABLE ***
MPU Bufferable Permission **ENABLE ***

Cortex Memory Protection Unit Region 2 Settings:

MPU Region **Enabled ***
MPU Region Base Address **0x30000000 ***

MPU Region Size	256KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	ENABLE *
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x38000000 *
MPU Region Size	64KB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	ENABLE *
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Enabled *
MPU Region Base Address	0x60000000 *
MPU Region Size	64MB *
MPU SubRegion Disable	0x0 *
MPU TEX field level	level 1 *
MPU Access Permission	ALL ACCESS PERMITTED *
MPU Instruction Access	DISABLE *
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	ENABLE *
MPU Bufferable Permission	ENABLE *

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled
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Cortex Memory Protection Unit Region 8 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

3.2. DEBUG

Debug: Serial Wire

3.3. ETH

Mode: RMII

3.3.1. Parameter Settings:

General : Ethernet Configuration:

Warning	The ETH can work only when RAM is pointing at 0x24000000
Ethernet MAC Address	00:80:E1:00:00:00
Tx Descriptor Length	4
First Tx Descriptor Address	0x30000080 *
Rx Descriptor Length	4
First Rx Descriptor Address	0x30000000 *
Rx Buffers Address	0x30000100 *
Rx Buffers Length	1536

3.4. FMC

SDRAM 1

Clock and chip enable: SDCKE0+SDNE0

Internal bank number: 4 banks

Address: 12 bits

Data: 32 bits

Byte enable: 32-bit byte enable

3.4.1. SDRAM 1:

SDRAM control:

Bank	SDRAM bank 1
Number of column address bits	8 bits
Number of row address bits	12 bits
CAS latency	2 memory clock cycles *
Write protection	Disabled
SDRAM common clock	2 HCLK clock cycles *
SDRAM common burst read	Disabled
SDRAM common read pipe delay	1 HCLK clock cycle *

SDRAM timing in memory clock cycles:

Load mode register to active delay	2 *
Exit self-refresh delay	7 *
Self-refresh time	4 *
SDRAM common row cycle delay	7 *
Write recovery time	3 *
SDRAM common row precharge delay	2 *
Row to column delay	2 *

3.4.2. Bank Mapping:

Mapping parameters:

FMC bank mapping	Default mapping
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3.5. I2C1

I2C: I2C

3.5.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
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I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x307075B1 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.6. I2C4

I2C: I2C

3.6.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	0x307075B1 *

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.7. LTDC

Display Type: RGB888 (24 bits)

3.7.1. Parameter Settings:

Synchronization for Width:

Horizontal Synchronization Width	39 *
Horizontal Back Porch	7
Active Width	800 *
Horizontal Front Porch	6
HSync Width	38
Accumulated Horizontal Back Porch Width	45
Accumulated Active Width	845
Total Width	851

Synchronization for Height:

Vertical Synchronization Height	3 *
Vertical Back Porch	2
Active Height	480
Vertical Front Porch	2
VSynC Height	2
Accumulated Vertical Back Porch Height	4
Accumulated Active Height	484
Total Height	486

Signal Polarity:

Horizontal Synchronization Polarity	Active Low
Vertical Synchronization Polarity	Active Low
Data Enable Polarity	Active Low
Pixel Clock Polarity	Normal Input

Layer Default Color:

Red	0
Green	0
Blue	0

3.7.2. Layer Settings:

Layer Default Color:

Layer 0 - Alpha	255 *
Layer 0 - Blue	0
Layer 0 - Green	0
Layer 0 - Red	0

Number of Layers:

Number of Layers	1 layer *
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Windows Position:

Layer 0 - Window Horizontal Start	0
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Layer 0 - Window Horizontal Stop	800 *
Layer 0 - Window Vertical Start	0
Layer 0 - Window Vertical Stop	480 *
Pixel Parameters:	
Layer 0 - Pixel Format	RGB888 *
Blending:	
Layer 0 - Alpha constant for blending	255 *
Layer 0 - Blending Factor1	Alpha constant x Pixel Alpha *
Layer 0 - Blending Factor2	Alpha constant x Pixel Alpha *
Frame Buffer:	
Layer 0 - Color Frame Buffer Start Address	0x60000000 *
Layer 0 - Color Frame Buffer Line Length (Image Width)	800 *
Layer 0 - Color Frame Buffer Number of Lines (Image Height)	480 *

3.8. MEMORYMAP

mode: Activated

3.9. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.9.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000
CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	4 WS (5 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 4 and 8 MHz
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PLL2 input frequency range	Between 4 and 8 MHz
PLL3 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range
PLL2 clock Output range	Wide VCO range
PLL3 clock Output range	Wide VCO range

3.10. SDMMC1

Mode: SD 4 bits Wide bus

3.10.1. Parameter Settings:

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.11. SYS

Timebase Source: SysTick

3.12. TIM5

Clock Source : Internal Clock

3.12.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

3.13. UART7

Mode: Asynchronous

3.13.1. Parameter Settings:

Basic Parameters:

Baud Rate	2000000 *
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.14. USART6

Mode: Asynchronous

3.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.15. FATFS

mode: SD Card

3.15.1. Set Defines:

Version:

FATFS version	R0.12c
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Function Parameters:

FS_READONLY (Read-only mode)	Disabled
FS_MINIMIZE (Minimization level)	Disabled
USE_STRFUNC (String functions)	Enabled with LF -> CRLF conversion
USE_FIND (Find functions)	Enabled *
USE_MKFS (Make filesystem function)	Enabled
USE_FASTSEEK (Fast seek function)	Enabled
USE_EXPAND (Use f_expand function)	Enabled *
USE_CHMOD (Change attributes function)	Enabled *
USE_LABEL (Volume label functions)	Disabled
USE_FORWARD (Forward function)	Disabled

Locale and Namespace Parameters:

CODE_PAGE (Code page on target)	Baltic *
USE_LFN (Use Long Filename)	Enabled with dynamic working buffer on the HEAP *
MAX_LFN (Max Long Filename)	255
LFN_UNICODE (Enable Unicode)	ANSI/OEM
STRF_ENCODE (Character encoding)	UTF-8
FS_RPATH (Relative Path)	Enabled with f_getcwd *

Physical Drive Parameters:

VOLUMES (Logical drives)	1
MAX_SS (Maximum Sector Size)	512
MIN_SS (Minimum Sector Size)	512
MULTI_PARTITION (Volume partitions feature)	Disabled
USE_TRIM (Erase feature)	Disabled
FS_NOFSINFO (Force full FAT scan)	0

System Parameters:

FS_TINY (Tiny mode)	Disabled
FS_EXFAT (Support of exFAT file system)	Disabled
FS_NORTC (Timestamp feature)	Dynamic timestamp
FS_REENTRANT (Re-Entrancy)	Enabled
FS_TIMEOUT (Timeout ticks)	2000 *
USE_MUTEX	Disabled
SYNC_t (O/S sync object)	osSemaphoreId
FS_LOCK (Number of files opened simultaneously)	10 *

3.15.2. Advanced Settings:

SDIO/SDMMC:

SDMMC instance	SDMMC1
Use dma template	Enabled
BSP code for SD	Generic

3.16. FREERTOS

Interface: CMSIS_V1

3.16.1. Config parameters:

API:

FreeRTOS API	CMSIS v1
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Versions:

FreeRTOS version	10.3.1
CMSIS-RTOS version	1.02

MPU/FPU:

ENABLE_MPU	Disabled
ENABLE_FPU	Enabled *

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock

TICK_RATE_HZ	1000
MAX_PRIORITIES	9 *
MINIMAL_STACK_SIZE	256 *
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Disabled
USE_COUNTING_SEMAPHORES	Disabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Enabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	131072 *
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Enabled *
USE_TICK_HOOK	Enabled *
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled *
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	512

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

3.16.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled *
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Disabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Disabled
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Disabled
xEventGroupSetBitFromISR	Enabled *
xTimerPendFunctionCall	Enabled *
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.16.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	Disabled
----------------------	----------

Project settings (see parameter description first):

Use FW pack heap file	Enabled
-----------------------	---------

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
ETH	PC1	ETH_MDC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA1	ETH_REF_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA2	ETH_MDIO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA7	ETH_CRSDV	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC4	ETH_RXD0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC5	ETH_RXD1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB11	ETH_TX_EN	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB12	ETH_TXD0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB13	ETH_TXD1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
FMC	PI9	FMC_D30	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI10	FMC_D31	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF0	FMC_A0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF1	FMC_A1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF2	FMC_A2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF3	FMC_A3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF4	FMC_A4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF5	FMC_A5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH2	FMC_SDCKE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH3	FMC_SDNE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH5	FMC_SDNWE	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF11	FMC_SDNRAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF12	FMC_A6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF13	FMC_A7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF14	FMC_A8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF15	FMC_A9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG0	FMC_A10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG1	FMC_A11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE7	FMC_D4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE8	FMC_D5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE9	FMC_D6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE10	FMC_D7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PE11	FMC_D8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE12	FMC_D9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE13	FMC_D10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE14	FMC_D11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE15	FMC_D12	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH8	FMC_D16	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH9	FMC_D17	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH10	FMC_D18	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH11	FMC_D19	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH12	FMC_D20	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD8	FMC_D13	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD9	FMC_D14	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD10	FMC_D15	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD14	FMC_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD15	FMC_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG4	FMC_BA0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG5	FMC_BA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG8	FMC_SDCLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH13	FMC_D21	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH14	FMC_D22	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH15	FMC_D23	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI0	FMC_D24	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI1	FMC_D25	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI2	FMC_D26	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI3	FMC_D27	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD0	FMC_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD1	FMC_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG15	FMC_SDNCAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE0	FMC_NBL0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE1	FMC_NBL1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI4	FMC_NBL2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI5	FMC_NBL3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI6	FMC_D28	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PI7	FMC_D29	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
I2C1	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB8	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
I2C4	PD12	I2C4_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PD13	I2C4_SDA	Alternate Function Open	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
			Drain			
LTDC	PE4	LTDC_B0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE5	LTDC_G0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE6	LTDC_G1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI11	LTDC_G6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI12	LTDC_HSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI13	LTDC_VSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI14	LTDC_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF10	LTDC_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC0	LTDC_R5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH4	LTDC_G5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA3	LTDC_B5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA5	LTDC_R4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA6	LTDC_G2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB0	LTDC_R3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB1	LTDC_R6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PI15	LTDC_R0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ0	LTDC_R1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ1	LTDC_R2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB10	LTDC_G4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG6	LTDC_R7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA8	LTDC_B3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	LTDC_B4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD3	LTDC_G7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD6	LTDC_B2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ12	LTDC_G3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PJ13	LTDC_B1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PK5	LTDC_B6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PK6	LTDC_B7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SDMMC1	PC8	SDMMC1_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC9	SDMMC1_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC10	SDMMC1_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC11	SDMMC1_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC12	SDMMC1_CK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD2	SDMMC1_CMD	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
UART7	PF6	UART7_RX	Alternate Function Push Pull	Pull-up *	Low	
	PF7	UART7_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART6	PC6	USART6_TX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
	PC7	USART6_RX	Alternate Function Push Pull	No pull-up and no pull-down	High *	
GPIO	PA15 (JTDI)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PB4 (NJTRST)	GPIO_EXTI4	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	

4.2. DMA configuration

DMA request	Stream	Direction	Priority
USART6_RX	DMA1_Stream0	Peripheral To Memory	Low
USART6_TX	DMA1_Stream1	Memory To Peripheral	Low
UART7_RX	DMA1_Stream2	Peripheral To Memory	Low
UART7_TX	DMA1_Stream3	Memory To Peripheral	Low

USART6_RX: DMA1_Stream0 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

USART6_TX: DMA1_Stream1 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

UART7_RX: DMA1_Stream2 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

UART7_TX: DMA1_Stream3 DMA request Settings:

Mode: Normal
 Use fifo: Disable

Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

MDMA Request	Channel	Trigger Mode	Priotity
SDMMC1_END_DATA	MDMA_Channel0	Buffer transfer	Low

SDMMC1_END_DATA: MDMA_Channel0 DMA request Settings:

Buffer Transfer Length: **16 ***
Data Mask: 0
Source Burst Size: Single transfer
Destination Burst Size: Single transfer
Block Data Length: 0
Destination Address: 0
Destination Block Address 0
Offset:
Address Mask: 0
Data Alignment: N/A
Source Block Address 0
Offset:
Endianness: Little endianness preserved
Source Address: 0
Source Increment mode: Increment by a byte
Block Count: 0
Destination Increment mode: Increment by a byte
Source Data Width: Byte
Destination Data Width: Byte

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
DMA1 stream0 global interrupt	true	5	0
DMA1 stream1 global interrupt	true	5	0
DMA1 stream2 global interrupt	true	5	0
DMA1 stream3 global interrupt	true	5	0
SDMMC1 global interrupt	true	5	0
USART6 global interrupt	true	5	0
UART7 global interrupt	true	5	0
MDMA global interrupt	true	5	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
EXTI line4 interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
FMC global interrupt	unused		
TIM5 global interrupt	unused		
Ethernet global interrupt	unused		
Ethernet wake-up interrupt through EXTI line 86	unused		
FPU global interrupt	unused		
LTDC global interrupt	unused		
LTDC global error interrupt	unused		
I2C4 event interrupt	unused		
I2C4 error interrupt	unused		
HSEM1 global interrupt	unused		

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	true	true
DMA1 stream0 global interrupt	false	true	true
DMA1 stream1 global interrupt	false	true	true
DMA1 stream2 global interrupt	false	true	true
DMA1 stream3 global interrupt	false	true	true
SDMMC1 global interrupt	false	true	true
USART6 global interrupt	false	true	true
UART7 global interrupt	false	true	true
MDMA global interrupt	false	true	true

* User modified value

5. System Views

5.1. Category view

5.1.1. Current

Category view

Power Domain view

Choose filters ...

... by Power Domain

D1

D2

D3

None

Middleware

FATFS

FREERTOS

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Other
BDMA		TIM5	ETH	LTDC			DEBUG		
CORTEX_M7			FMC						
DMA			I2C1						
GPIO			I2C4						
MDMA			SDMMC1						
IVIC			UART7						
RCC			USART6						
SYS									

5.1.2. Without filters

Category view

Power Domain view

Choose filters ...

... by Power Domain

D1

D2

D3

None

Middleware

FATFS

FREERTOS

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal	Other
BDMA		TIM5	ETH	LTDC			DEBUG		
CORTEX_M7			FMC						
DMA			I2C1						
GPIO			I2C4						
MDMA			SDMMC1						
IVIC			UART7						
RCC			USART6						
SYS									

5.2. Power Domain view

Category view

Power Domain view

D1

CORTEX_M7

FMC

LTDC

MDMA

HVIC

SDMMC1

D2

DMA

ETH

I2C1

TIM5

UART7

USART6

D3

BDMA

I2C4

RCC

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdI.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flpowerstbd.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-uart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4539-hrtim-cookbook-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4936-migration-of-microcontroller-applications-from-stm32f7-series-to-stm32h743753-line-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f477h7-series-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5312-migration-from-revy-to-revv-for-stm32h743753-and-stm32h750-value-line-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5354-getting-started-with-the-stm32h7-series-mcu-16bit-adc-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an5342--how-to-use-error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf

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